

Mathematical Induction

$$n \rightarrow n+1$$

Mathematical Induction

$P(n)$: Mathematical Statement depending on $n \in \mathbb{N}$.

Proposition: $\forall n \in \mathbb{N} \ P(n)$ is true.

proof: • Show $P(1)$ is true (base case)

• Prove for arbitrary $k \in \mathbb{N}$

$$P(k) \Rightarrow P(k+1) \quad (\text{induction step})$$

$$\bullet P(1) \Rightarrow P(2) \Rightarrow P(3) \Rightarrow \dots$$

Proposition: $\forall n \in \mathbb{N} \ 1+3+5+\dots+(2n-1) = n^2$

proof: Base ($P(1)$) Clearly $1=1^2 \quad \checkmark$

Suppose $k \geq 1$ holds $1+3+\dots+(2k-1) = k^2$

Consider $1+3+5+\dots+(2k-1)+(2k+1)$

$$= k^2 + 2k+1 = (k+1)^2 \quad \checkmark$$

By the PMI we have $1+3+\dots+(2n-1) = n^2 \quad \forall n \in \mathbb{N}$ $\square$

Proposition: $\forall 2n-1 \in \mathbb{N}$, we have $3 \mid 2^n + 1$.

proof: Base case Observation $3 = 3 = 2^1 + 1 \Rightarrow 3 | 2^1 + 1$

Suppose that for some odd $k \geq 1$ we have $3 | 2^k + 1$
 $\Rightarrow 2^k + 1 = 3m$ for some $m \in \mathbb{N}$.

The next odd number is $k+2$: $2^{k+2} + 1 = 4 \cdot 2^k + 1$
 $= 4 \cdot 2^k + 1 + 4 - 4 = 4(2^k + 1) - 3 = 4 \cdot 3m - 3$
 $= 3(4m - 1) \Rightarrow 3 | 2^{k+2} + 1 \quad \square$

Proposition: $\forall n \in \mathbb{N} \quad 6 | 17n^3 + 103n$

proof: Base case ($n=1$) $17n^3 + 103n = 120 = 6 \cdot 20 \quad \checkmark$

Suppose for some $k \geq 1$ $6 | 17k^3 + 103k$.

So $17k^3 + 103k = 6m$ for some $m \in \mathbb{N}$.

Consider $17(k+1)^3 + 103(k+1) = 17(k^3 + 3k^2 + 3k + 1) + 103k + 103$

$$= 17k^3 + 103k + 120 + 17(3k^2 + 3k)$$

$$= 6(m+20) + 17 \cdot 3k(k+1) = 6M.$$

$\uparrow$
one of those is even $\square$

Proposition: If $f_1 = f_2 = 1$, $f_{n+2} = f_{n+1} + f_n$ then

$$f_1 + f_2 + \dots + f_n = f_{n+2} - 1$$

proof: Base case: Notice $f_3 = f_2 + f_1 = 2$.

$$f_1 = 1 = 2 - 1 = f_3 - 1. \quad \checkmark$$

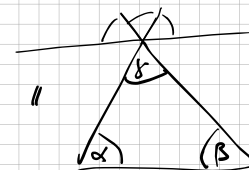
Suppose for $k \geq 1$ $f_1 + \dots + f_k = f_{k+2} - 1$.

Consider $f_1 + \dots + f_k + f_{k+1} = f_{k+2} - 1 + f_{k+1}$
 $= f_{k+3} - 1 \quad \square$

Proposition: For $n \geq 3$, the sum of the interior angles of a convex n -gon is $(n-2)\pi$.

proof: Base case: ($n=3$)

$$\Rightarrow \alpha + \beta + \gamma = \pi = (3-2)\pi$$



Suppose for $k \geq 3$ the sum of angles of a k -gon is $(k-2)\pi$.

Consider a $(k+1)$ -gon.



$$\begin{aligned} \text{sum of angles of } (k+1)\text{-gon} &= \text{sum of } k\text{-gon} + \text{sum of } \triangle A_1 A_k A_{k+1} \\ &= (k-2)\pi + \pi = (k-1)\pi \quad \square \end{aligned}$$